

BERT'S

Sheet Metal Products Inc.

Laser Air Compressor Maintenance

Guide

Hoverr / Haoweier permanent-magnet variable-speed screw compressor with tank, dryer and filters. What to do daily, weekly, at 500 hours and every 3,000 hours, and how to clear the service alarms.

Bert's Sheet Metal Products, Inc. · 9521 Irondale Ave, Chatsworth, CA 91311 · 818.775.0104 · BSMP-MG-001 Rev A

Right now: the GREASE service alarm

The controller keeps five service-hour counters (air filter, oil filter, oil-air separator, lube oil, grease). When a counter reaches its preset limit the screen raises a maintenance warning for that item. The compressor keeps running on a warning; it is a reminder, not a fault. A **GREASE** warning means the grease counter has reached its limit (factory preset 3,000 hours).

What the manual does and does not say
The manual shows the GREASE counter and how to reset it, but it gives **no grease point, quantity or grease type** anywhere in its 41 pages. On this class of machine the grease interval is for the **permanent-magnet motor bearings**. The motor on our unit is a fully enclosed can with a fan grille on the back (see photo on page 3). Before anyone puts a grease gun on it, confirm with the supplier whether our motor has grease fittings at all; many of these motors use sealed bearings and the counter is only a reminder to inspect.

Do this, in order

1. Write down the hours. Home screen, bottom left: **TOTAL TIME**. Then Menu → Consumable Parameters: note all five RUN TIME values. Put them in the service log on the last page.
2. Call or message the supplier (contact on page 4). Ask three things: (a) is the grease alarm the motor bearings, and where are the fittings; (b) what grease and how much per fitting; (c) the consumable-parameters password so we can reset timers ourselves. Ask for it in writing so it goes in this binder.
3. If the supplier confirms grease fittings: stop the unit, isolate power, wait for the VFD charge light to go out, then add the specified grease sparingly (a few strokes). Over-greasing a motor bearing does more harm than under-greasing. Wipe the excess. Never mix grease types.
4. If the supplier says sealed bearings: open the rear panel, blow the motor intake vents and grille clean from the outside in with shop air, listen for bearing noise on restart, and log it as “inspected, no fittings”.
5. Reset the counter: Menu → Consumable Parameters → enter password → set **GREASE RUN TIME** to 0 → check **GREASE MAX TIME** still reads 3000 → Menu. On the home screen press and hold **RESET** for 3 seconds to clear the warning.
6. While the panels are open, check the other four counters against their limits. If any are close, do that service now rather than opening the machine twice.

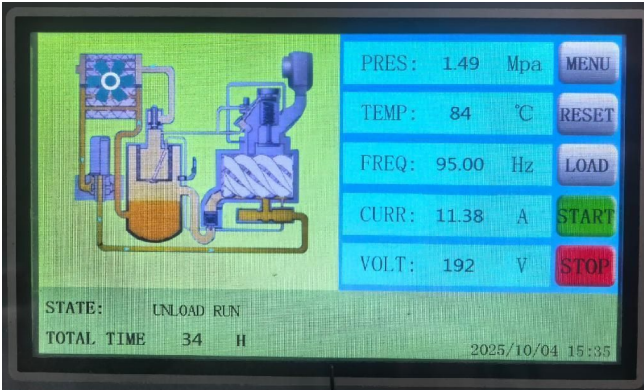
Counters and what they mean

Counter on screen	Service item	Factory limit	Then
AIR RUN TIME	Intake air filter element	500 h (first)	3,000 h or 6 months
OIL RUN TIME	Oil filter element (spin-on)	500 h (first)	3,000 h or 6 months
O-A RUN TIME	Oil-air separator element (inside the tank)	3,000 h	3,000 h or 6 months
LUBE RUN TIME	Screw compressor oil (change)	500 h (first)	3,000 h or 6 months, sooner if oil looks dark or milky
GREASE RUN TIME	Grease (motor bearings; confirm with supplier)	3,000 h	3,000 h

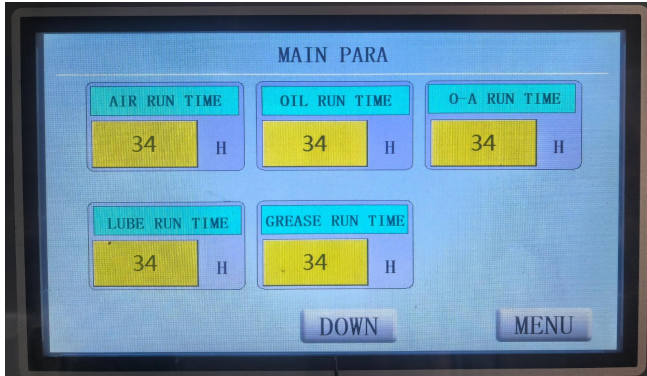
After the first 500-hour service the manual says to set every MAX TIME to 3000 hours and every RUN TIME to 0. “Whichever comes first” is the rule: at our duty cycle six months will usually arrive before 3,000 hours.

The controller: where the timers live

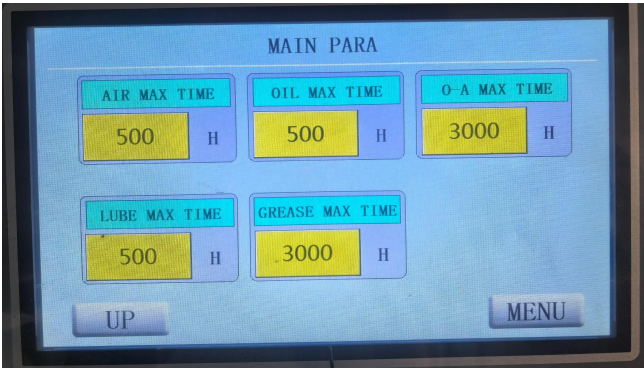
The touch screen has a home page and a menu. **Consumable Parameters** is the only page you need for service alarms. Do not go into Factory Parameters, VFD Parameters or Fan Settings; those are preset and a wrong entry can damage the unit.



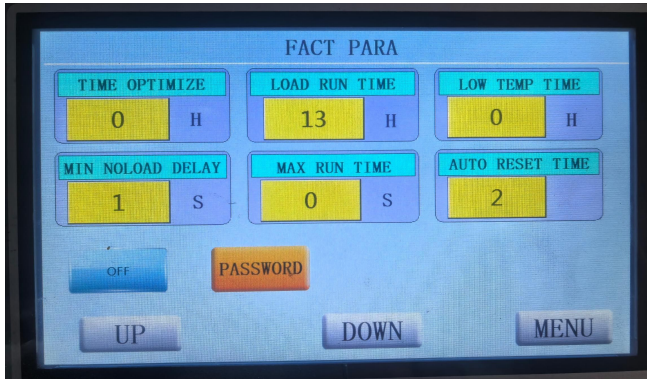
Home screen. TOTAL TIME bottom left. RESET clears a warning (hold 3 s). START is hold 3 s; STOP is one tap.



Consumable Parameters, page 1: hours run since last reset on each item. The alarm fires when RUN TIME reaches MAX TIME.



Consumable Parameters, page 2 (DOWN): the limits. 500 h on air, oil and lube for a new machine; 3,000 h on separator and grease.



Factory Parameters (password). LOAD RUN TIME here is loaded hours; the manual's reset page sits behind the same PASSWORD button.

Home-screen buttons

Button	What it does
MENU	Opens the secondary menu: Operating Parameters, User Parameters, Factory Parameters, Consumable Parameters, Fault Log.
RESET	Press and hold 3 seconds after a warning or fault has been dealt with. Clears the message.
LOAD	Manual mode only: tap to load, tap again to unload.
START	Press and hold 3 seconds to start.
STOP	Tap once. Normal shutdown (unload, then stop after the shutdown delay). Use the red mushroom E-stop only in an emergency.

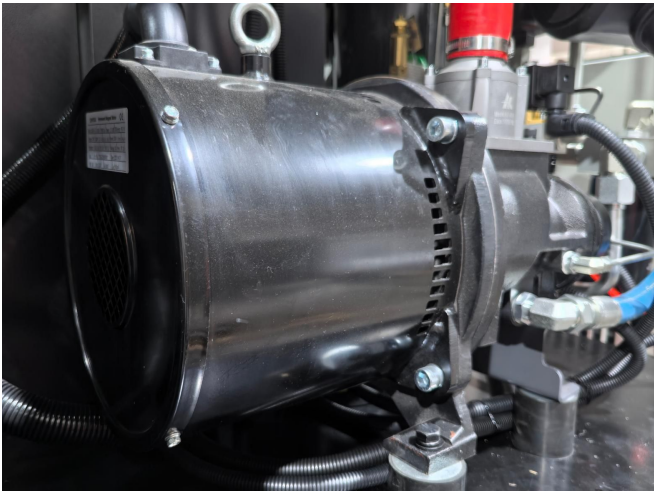
Reading the home screen while it runs

- **PRES** in MPa. 1.6 MPa = 232 psi is the working pressure. Load and unload setpoints live in User Parameters.
- **TEMP** is discharge temperature. Normal is roughly 75 to 95 °C. Warning at 105 °C, trip at 110 °C. A rising trend means a dirty cooler, low oil, or a room over 40 °C.
- **FREQ** and **CURR** show how hard the VFD is working. Running at full frequency all day means the laser is drawing more air than the unit makes, or there is a leak.
- **STATE** reads LOAD RUN, UNLOAD RUN or STOP. Long stretches of UNLOAD RUN waste power; the no-load delay in User Parameters (90 to 360 s) controls how long it idles before stopping.

Before opening the cabinet

- Every time, no exceptions**
1. Press STOP and let it wind down.
 2. Open the disconnect and lock it.
 3. Wait until the VFD charge light is out; the drive holds a lethal charge after power-off.
 4. Vent the system to 0 on the internal pressure gauge before touching any filter housing, the separator cover or the pressure sensor.
 5. Power off the electronic drain valve before pulling the multi-stage filter housings.
 6. Cooler fins, the airend and the motor are hot for a long time after a run.
 7. No watches or rings near the drive; insulated tools only. Only someone trained on this unit does electrical work.

What is inside



Permanent-magnet motor coupled straight to the airend. The grille on the end is the motor's own cooling intake: blow it clean weekly.



Oil-air separator tank cover with the minimum-pressure valve on top. The separator element is under this plate; the oil fill port and drain are on the tank.

Part	Where	Why you care
Intake air filter (paper element)	Behind the intake grille at the back	Dirty element starves the airend and drives up temperature
Oil filter (spin-on can)	On the airend / oil circuit	Changed at 500 h then every 3,000 h
Oil sight glass and fill port	On the separator tank	Idle oil level must sit between the two red lines
Oil-air separator element	Inside the tank, under the bolted cover	Worn element = oil carry-over into the laser air
Minimum-pressure valve, safety valve	Top of the separator tank	Do not adjust
Pressure sensor	On the discharge line	Condensate in its probe hole gives false pressure readings (page 5)
Refrigerated dryer	Separate box on the tank	Must run whenever the compressor runs; evaporator gauge on its front
Multi-stage filters (AO / AA / AX / AAR / ACS)	Row of housings after the dryer	The laser's clean-air guarantee; change as a set every 3,000 h or 6 months
Primary filter cartridge + electronic drain	Base of the first housing	Drain must be powered off before the housing comes apart
Air receiver manual drain	Bottom of the tank	Open 2 to 3 times a day

Supplier

Taike Machinery Technology (Dongguan) Co., Ltd, brand names Hoverr and Haoweier. Contact printed in the manual: Meng Hui, mobile +86 136 0266 7790 (WeChat works on that number). Have the model and the hour reading ready. The consumable-parameters password and the grease specification both have to come from them; write the answers on the log page.

Our model

The manual covers the whole range. Circle ours from the nameplate on the front panel so the right parts list on page 6 gets used:

Model	Motor	Air delivery	Laser it is sized for	Tank	Parts list
HB11PM-16-300	11 kW	1.0 m³/min	1 to 3 kW	350 L	List A
HB15PM-16-300	15 kW	1.35 m³/min	4 to 6 kW	350 L	List A
HB22PM-16-300	22 kW	2.0 m³/min	6 to 12 kW	single	List B
TGJ-30VSD-16-500	22 kW	2.0 m³/min	10 to 15 kW	twin 560 L	List C
TGJ-50VSD-16-500	37 kW	3.7 m³/min	15 to 20 kW	twin 560 L	List D
TGX-50VSD-16-2600	37 kW	3.7 m³/min	22 to 50 kW	dual 600 L skid	List E

All models: 1.6 MPa (232 psi) working pressure, air cooled, 380 V 3-phase (dryer on 220 V), continuous duty. Ambient 10 to 40 °C, at least 70 cm clear on every side.

User Parameters you are allowed to touch

Setting	What it is	Manual's guidance
Load / Unload pressure	Where the compressor starts and stops making air	Set for the laser; keep unload below the 1.6 MPa high limit
No-load delay	Idle time before it shuts down after unloading	90 to 360 seconds
Restart delay	Wait before it may restart after a stop	5 to 10 seconds
Shutdown delay	Wind-down after STOP is pressed	10 to 30 seconds
Language	Toggles English / Chinese	One tap

Service schedule

When	Do	Who
2 to 3 times a day	Open the manual drain on the air receiver until it blows dry air. Glance at PRES and TEMP on the screen.	Laser operator
Weekly	Blow out the intake filter screen from the inside out. Blow the motor's side intake vents inward. Blow the cooler fins clean. Wipe dust off the VFD and inside the electrical box (power off). Check the electronic drain valve actually cycles. Check the oil level in the sight glass with the unit stopped.	Laser operator
First 500 hours (new machine)	Change the air filter, the oil filter and the oil. Then set the three 500-hour limits to 3000 and zero the counters.	Manager + supplier password
Every 3,000 hours or 6 months, whichever first	Air filter, oil filter, oil, oil-air separator element, every multi-stage filter element, primary filter cartridge. Grease per supplier. Zero all counters.	Manager
Any time TEMP creeps up	Cooler fins, intake filter, oil level, room temperature, in that order.	Anyone
Any time the pressure reading looks wrong	Pressure-sensor condensate procedure on this page.	Manager

Oil

- Use only the supplier's screw-compressor oil (kit item 11020004, 10 L bucket; 22 kW units take one, 37 kW take two). **Never mix brands.** If the oil type is unknown, drain and refill, do not top up.
- Fill only to the **upper red line** on the sight glass. Overfilling pushes oil into the air and into the laser.
- Check the level with the unit stopped and pressure released. Running level reads low; that is normal.
- Waste oil comes out of the drain valve at the bottom of the separator tank.

Oil-air separator element (manual, page 25)

1. Stop, isolate, vent to 0. Drain the oil completely.
2. Disconnect the external pipes from the separator cover (oil return line and the minimum-pressure valve line).
3. Loosen all perimeter bolts and lift the cover. Pull the old element.
4. Look inside the tank for sludge or rust; clean if needed.
5. Fit the new element. Keep the staple on the gasket (it grounds the element against static). Smear a thin film of oil on the gasket.
6. Refit the cover, align the holes, tighten the bolts diagonally and evenly. Reconnect the pipes. Refill oil to the upper red line.

Multi-stage filters and primary cartridge (manual, page 26)

1. Vent to 0. Power off the electronic drain valve. Remove the drain valve from the base of the first housing.
2. Unscrew each bowl, swap the element, check the seal ring is seated, refit.
3. Keep the grades in order (AO, then AA, then AX and so on). Never put a fine element where a coarse one goes.
4. Filters made before May 2023 used the 034 / 015 series; later units use 035 / 018. **Read the label on the lower housing before ordering.**

Pressure reads wrong or the safety valve pops (manual, page 28)

Cause: condensate collects in the probe hole under the pressure sensor when the room is humid, the machine stopped hot, or it sat idle. Fix: stop, isolate, vent to 0. Unscrew the sensor, blow the hole at its base dry with an air gun, refit, restore power, restart.

Air filter and oil filter

Air element: open the back panel, unclip the housing, swap the paper element, refit. Oil filter: spin off the old can, wet the new gasket with oil, spin on hand-tight plus a quarter turn. Both are 500 hours on a new machine, then 3,000 hours or 6 months.

Parts lists (from the manual, pages 31 to 35)

Quote the material code when ordering. Every list: first service at 500 h is air filter, oil filter and oil; after that everything below every 3,000 h or 6 months.

List A: HB11PM-16-300 / HB15PM-16-300 (11 to 15 kW)

Code	Spec	Item	Qty	Note
68082909	11-15KW	Air filter	1	500 h first, then 3,000 h
68083506	11-15KW	Oil filter	1	500 h first, then 3,000 h
11020004	10 L	Screw compressor oil	1	500 h first, then 3,000 h
68083501	11-15KW	Oil-gas separator	1	
10020401 / 402 / 403	GNY-015-AO / AA / AX	Filter elements	1 ea	Units built before May 2023
10020421 / 422 / 423	GNY-018-AO / AA / AX	Filter elements	1 ea	Units built from June 2023
13020370	11-15KW	Primary filter cartridge	1	Old and new styles exist; check the housing label

List B: HB22PM-16-300 (22 kW, single tank)

Code	Spec	Item	Qty	Note
68082910	22-37KW	Air filter	1	500 h first, then 3,000 h
68083504	22-37KW	Oil filter	1	500 h first, then 3,000 h
11020004	10 L bucket	Screw compressor oil	1	500 h first, then 3,000 h
68083502	22KW built-in	Oil-gas separator	1	
10020431 / 432 / 433	GNY-034-AO / AA / AX	Filter elements	1 ea	Units built before May 2023
10020441 / 442 / 443	GNY-035-AO / AA / AX	Filter elements	1 ea	Units built from June 2023
13020371	22KW	Primary filter element	1	

List C: TGJ-30VSD-16-500 (22 kW, twin tank)

Code	Spec	Item	Qty	Note
68082910	22-37KW	Air filter	1	500 h first, then 3,000 h
68083504	22-37KW	Oil filter	1	500 h first, then 3,000 h
11020004	10 L bucket	Screw compressor oil	1	500 h first, then 3,000 h
68083502	22KW built-in	Oil-gas separator	1	
10020441	GNY-035-AO	Filter element	1	
10020442	GNY-035-AA	Filter element	1	
10020443	GNY-035-AX	Filter element	2	
10020445	GNY-035-AAR	Filter element	1	
10020444	GNY-035-ACS	Filter element	1	
13020371	22-37KW	Primary filter element	1	
12053208	GNY-16KG	Electronic drain valve	1	Spare

List D: TGJ-50VSD-16-500 (37 kW, twin tank)

Code	Spec	Item	Qty	Note
68082910	22-37KW	Air filter	1	500 h first, then 3,000 h
68083504	22-37KW	Oil filter	1	500 h first, then 3,000 h
11020004	10 L bucket	Screw compressor oil	2	500 h first, then 3,000 h
13020374	37KW built-in	Oil-gas separator	1	
10020441	GNY-035-AO	Filter element	1	
10020442	GNY-035-AA	Filter element	1	
10020443	GNY-035-AX	Filter element	2	
10020445	GNY-035-AAR	Filter element	1	
10020444	GNY-035-ACS	Filter element	1	
13020371	22-37KW	Primary filter element	1	
12053208	GNY-16KG	Electronic drain valve	1	Spare

List E: TGX-50VSD-16-2600 (37 kW, skid)

Code	Spec	Item	Qty	Note
68082910	22-37KW	Air filter	1	500 h first, then 3,000 h
68083504	22-37KW	Oil filter	1	500 h first, then 3,000 h
11020004	10 L bucket	Screw compressor oil	2	500 h first, then 3,000 h
13020374	37KW built-in	Oil-gas separator	1	
10020446 / 447	GNY-065-AO / AA	Filter elements	1 ea	
10020448	GNY-065-AX	Filter element	2	
10020449 / 450	GNY-065-AAR / ACS	Filter elements	1 ea	
13020371	22-37KW	Primary filter element	1	
09080404 / 405 / 406	GNY-CM5.0 /A /T /H	High-precision filter sets	1 ea	Optional high-precision air option
09120110 + 10080111	GNY-026-AX	Cutting-head protection filter + element	1	Optional
12053208	GNY-16KG	Electronic drain valve	1	Spare

Service log

One line per visit. Hours are the TOTAL TIME on the home screen. Tick the counters you zeroed. Keep this page in the binder by the laser; photograph it into the QC app when it fills.

Supplier answers (fill in once)	
Consumable-parameters password	
Grease point(s) and quantity	
Grease type / part number	
Oil type / part number	11020004 (10 L)
Our model and serial	
Date installed / first start	

Air / Oil / O-A / Lube / Grease columns: tick when that counter was zeroed on the controller at this visit.

Date	Total hours	Work done	Air	Oil	O-A	Lube	Grease	By